

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			At least 2 wavefronts with a smaller wavelength (1) At least 1 wavefront joined to those in deep water (1) N.B. Refracted rays must be going in the correct direction to award any marks i.e. bottom RH corner		2		2		
	(b)			Selection of: $v = f\lambda$ (1) Wavelength measured to be 1.2 [cm] or 12 [mm] or frequency = 0.5 [Hz] (1) $v = 0.5 \times 0.012 = 0.006$ [m/s] (1) cao Alternative: Selection of: speed = $\frac{\text{distance}}{\text{time}}$ (1) Wavelength measured to be 1.2 [cm] or 12 [mm] or distance = 6 [cm] or 60 [mm] (1) speed = $\frac{0.060}{10} = 0.006$ [m/s] (1) cao	1	1 1		3	2	
	(c)			Frequency same in both regions (1) Wavelength longer in deep / shorter in shallow (1) Wave speed greater in deep / smaller in shallow (1)	3			3		
				Question 2 total	4	4	0	8	2	0